

Customer Segmentation Using RFM Analysis and K-Means Clustering

Project Overview

Customer retention is often more cost-effective than acquiring new customers. The goal of this project was to identify meaningful customer segments using purchasing behavior and develop actionable business recommendations to improve customer retention and marketing effectiveness.

Using transactional retail data, customers were evaluated based on Recency, Frequency, and Monetary (RFM) metrics. These features were used to group customers into segments with similar purchasing behaviors using K-Means clustering.

Business Objective

The objective of this project was to identify valuable customer segments, recognize customers at risk of churn, and provide data-driven recommendations that could improve customer engagement and increase revenue.

Dataset

The dataset contains online retail transaction records, including customer purchases, invoice information, product details, quantities, prices, and purchase dates.

Tools and Technologies

- Python
- Jupyter Notebook
- pandas
- NumPy
- scikit-learn
- Matplotlib
- SQL (data exploration)
- Tableau (visualization)

Data Preparation

Several preprocessing steps were completed before modeling:

- Removed canceled transactions.

- Handled missing customer identifiers.
- Removed invalid or negative values.
- Created Total Sales values.
- Calculated Recency, Frequency, and Monetary (RFM) metrics for each customer.
- Standardized features before clustering.

Methodology

RFM analysis was used to summarize customer purchasing behavior.

- Recency measured how recently a customer made a purchase.
- Frequency measured how often customers purchased.
- Monetary measured total customer spending.

K-Means clustering was then applied to identify groups of customers with similar purchasing patterns.

Results

The analysis successfully identified distinct customer groups, including high-value customers, loyal customers, average customers, and customers at risk of churn.

These segments provide valuable information for targeted marketing campaigns, customer retention strategies, and personalized promotions.

Business Recommendations

- Reward high-value customers with loyalty incentives.
- Develop personalized campaigns for average customers.
- Target at-risk customers with retention offers before they disengage.
- Continue monitoring customer behavior to identify changes over time.

Skills Demonstrated

- Data cleaning
- Exploratory Data Analysis (EDA)
- Feature engineering
- Customer analytics

- RFM segmentation
- K-Means clustering
- Business communication
- Data visualization
- Predictive analytics concepts

Future Improvements

Future work could include integrating demographic information, evaluating alternative clustering algorithms, developing automated ETL pipelines, and incorporating predictive machine learning models for customer churn.

This is already at the level I'd expect from a junior data analyst portfolio.

The next thing I'd add is **screenshots** from your notebook. Recruiters often spend less than two minutes on a portfolio, so showing visuals immediately helps them understand your work. For this project, I'd include:

- Your RFM distribution charts.
- The K-Means cluster visualization.
- Any Tableau dashboard you created.
- A simple workflow diagram showing **Raw Data → Cleaning → RFM → K-Means → Business Recommendations**